

B-12

$$\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} R(r) + \frac{2\mu}{\hbar^2} \left(-\frac{l(l+1)}{2\mu r^2} + \frac{Ze^2}{r} - |E| \right) R(r) = 0$$

$$\Rightarrow \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} \frac{d}{dp} = \frac{d}{dr} \quad \frac{1}{r} = \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} \frac{1}{p}$$

$$\Rightarrow \left(\frac{8\mu E}{\hbar^2} \right) \left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) + \frac{2\mu}{\hbar^2} \left(-\frac{l(l+1)}{2\mu p^2} \left(\frac{8\mu E}{\hbar^2} \right) + \frac{Ze^2}{p} \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} - |E| \right) R(p) = 0$$

$$\frac{8\mu E}{\hbar^2} \left\{ \left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) - \frac{l(l+1)}{p^2} + \left(\frac{4}{\hbar^2} \right) \frac{Ze^2}{p} \left(\frac{\hbar^2}{8\mu E} \right)^{1/2} - \left(\frac{Ze^2}{\hbar^2} \right) \left(\frac{\hbar^2}{8\mu E} \right)^{1/2} \right\} R(p) = 0$$

$$\left\{ \left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) - \frac{l(l+1)}{p^2} + \frac{2Ze^2}{p\hbar^2} \left(\frac{\hbar^2}{8\mu E} \right)^{1/2} - \frac{1}{4} \right\} R(p) = 0$$

$$\Rightarrow \left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) - \frac{l(l+1)}{p^2} R(p) + \frac{\lambda}{p} R(p) - \frac{1}{4} R(p) = 0$$

Asympt. behavior $\approx V^2 - \frac{1}{4} = 0 \Rightarrow V = \pm \frac{1}{2} \Rightarrow R=0 \Rightarrow e^{\pm p/2} \quad V' = \mp \frac{1}{2} e^{\pm p/2} \quad V'' = \frac{1}{4} e^{\pm p/2}$

$$R(p) \approx e^{\pm p/2} \quad \lim_{p \rightarrow \infty} \Rightarrow R(p) \approx e^{-p/2} \Rightarrow R(p) = G(p) e^{-p/2}$$

$$R = G e^{-p/2}$$

$$R' = G' e^{-p/2} - \frac{1}{2} G e^{-p/2}$$

$$R'' = G'' e^{-p/2} - G' e^{-p/2} + \frac{1}{4} G e^{-p/2}$$

$$\Rightarrow \begin{cases} R = G \\ R' = G' - \frac{1}{2} G \\ R'' = G'' - G' + \frac{1}{4} G \end{cases}$$

$$\left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) - \frac{l(l+1)}{p^2} R(p) + \frac{\lambda}{p} R(p) - \frac{1}{4} R(p) = 0$$

$$G'' - G' + \frac{1}{4} G + \frac{2}{p} \left(G' - \frac{1}{2} G \right) - \frac{l(l+1)}{p^2} G + \frac{\lambda}{p} G - \frac{1}{4} G = 0$$

$$\Rightarrow G'' + G' \left(-1 + \frac{2}{p} \right) + G \left(\frac{1}{4} - \frac{1}{p} - \frac{l(l+1)}{p^2} + \frac{\lambda}{p} - \frac{1}{4} \right) = 0$$

$$\Rightarrow G'' + G' \left(\frac{2}{p} - 1 \right) + G \left(-\frac{l(l+1)}{p^2} + \frac{\lambda-1}{p} \right) = 0$$

$$G = p^l H \Rightarrow G' = l p^{l-1} H + p^l H'$$

$$G'' = l(l-1) p^{l-2} H + 2l p^{l-1} H' + p^l H''$$

$$\Rightarrow G = H \quad G' = \frac{l}{p} H + H' \quad G'' = H'' + \frac{2l}{p} H' + \frac{l(l-1)}{p^2} H$$

$$H'' + \frac{2l}{p} H' + \frac{l(l-1)}{p^2} H + \left(\frac{l}{p} H + H' \right) \left(\frac{2}{p} - 1 \right) + H \left(\frac{\lambda-1}{p} - \frac{l(l+1)}{p^2} \right) = 0$$

$$H'' + H' \left(\frac{2(l+1)}{p} - 1 \right) + H \left(\frac{l(l-1)}{p^2} + \frac{2l}{p^2} - \frac{l}{p} + \frac{\lambda-1}{p} - \frac{l(l+1)}{p^2} \right) = 0$$

$$\frac{l^2-l}{p^2} + \frac{2l}{p^2} - \frac{l}{p} + \frac{\lambda-1}{p} - \frac{l^2+l}{p^2} = \frac{\lambda-1-l}{p}$$

$$H'' + H' \left(\frac{2(l+1)}{p} - 1 \right) + H \left(\frac{\lambda-1-l}{p} \right) = 0$$

$$H = \sum_{n=0}^{\infty} a_n p^n \Rightarrow H' = \sum_{n=0}^{\infty} a_{n+1} p^n \Rightarrow H'' = \sum_{n=0}^{\infty} a_{n+2} p^n$$

$$\sum_{n=0}^{\infty} a_{n+2} p^{n+1} + \sum_{n=0}^{\infty} a_{n+1} p^n \left(\frac{2(l+1)}{p} - 1 \right) + \sum_{n=0}^{\infty} a_n p^n (\lambda-1-l) = 0$$

$$\Rightarrow \sum_{n+1=1}^{\infty} a_{n+1} (n+1) p^n + \sum_{n+1=1}^{\infty} a_{n+1} (n+1) p^{n-1} (2l+2) = - \sum_{n=0}^{\infty} a_n p^{n-1} (n+1+l+n)$$

$$a_{n+1} (n+1) (2(l+1)+n) = -(n+1+l+n) a_n \Rightarrow \frac{a_{n+1}}{a_n} = \frac{(n+1+l-n)}{(n+1)(2(l+1)+n)}$$

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$$\vec{\nabla} \cdot \vec{E} = \mu_0 \rho \quad \vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt} \quad \vec{\nabla} \cdot \vec{B} = 0 \quad \vec{\nabla} \times \vec{B} = \frac{\mu_0}{c} \vec{J} + \frac{d\vec{E}}{dt}$$

$$\vec{B} = -\vec{\nabla} \times \vec{A}$$

$$\vec{E} = -\vec{\nabla} \phi - \frac{d\vec{A}}{dt}$$

$$E = -\nabla \phi - \frac{dA}{dt}$$

$$\nabla \cdot (-\nabla \phi - \frac{dA}{dt}) = \mu_0 \rho \Rightarrow -\nabla^2 \phi - \frac{d}{dt} \nabla \cdot \vec{A} = \mu_0 \rho \Rightarrow -\nabla^2 \phi = \mu_0 \rho \quad (\text{Poisson Denk})$$

$$\vec{\nabla} \times \vec{B} = \frac{\mu_0}{c} \vec{J} + \frac{d\vec{E}}{dt} \Rightarrow \vec{\nabla} \times (-\vec{\nabla} \times \vec{A}) = \frac{\mu_0}{c} \vec{J} + \frac{d}{dt} (-\nabla \phi - \frac{dA}{dt})$$

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$$

$$\Rightarrow -\nabla(\nabla \cdot \vec{A}) + \nabla^2 \vec{A} + \frac{d}{dt} \nabla \phi + \frac{d^2 A}{dt^2} = \frac{\mu_0}{c} \vec{J} \Rightarrow$$

$$\Rightarrow \nabla(-\nabla \cdot \vec{A} + \frac{d}{dt} \phi) + \nabla^2 \vec{A} + \frac{d^2 A}{dt^2} = \frac{\mu_0}{c} \vec{J}$$

$$0 \Rightarrow \nabla \cdot \vec{A} = -\frac{d\phi}{dt} \Rightarrow \text{Lorentz şartı}$$

$$\nabla \cdot \vec{E} = \mu_0 \rho \Rightarrow \nabla \cdot (-\nabla \phi - \frac{dA}{dt}) = \mu_0 \rho$$

$$\Rightarrow -\nabla^2 \phi + \frac{d}{dt} \nabla \cdot \vec{A} = \mu_0 \rho \Rightarrow$$

$$J^\mu = (c\rho, \vec{J}) = \nabla^\mu, \square \in \text{De Lambert op}$$

$$H\psi = E\psi \Rightarrow \psi = ? \quad H = ? \Rightarrow H = \frac{p^2}{2m} + \frac{eA}{c} + \frac{e\phi}{c}$$

$$\Rightarrow \frac{1}{2m} p^2 + \frac{eA}{c} \psi + \frac{e\phi}{c} \psi = E\psi$$

$$\frac{1}{2m} (p + \frac{eA}{c})(p + \frac{eA}{c}) \psi = E\psi \Rightarrow \frac{1}{2m} (\frac{\hbar}{i} \vec{\nabla} + \frac{eA}{c}) (\frac{\hbar}{i} \vec{\nabla} + \frac{eA}{c}) \psi = E\psi$$

$$\Rightarrow \frac{1}{2m} (\frac{\hbar}{i} \vec{\nabla} + \frac{eA}{c}) (\frac{\hbar}{i} \vec{\nabla} \psi + \frac{eA}{c} \psi) =$$

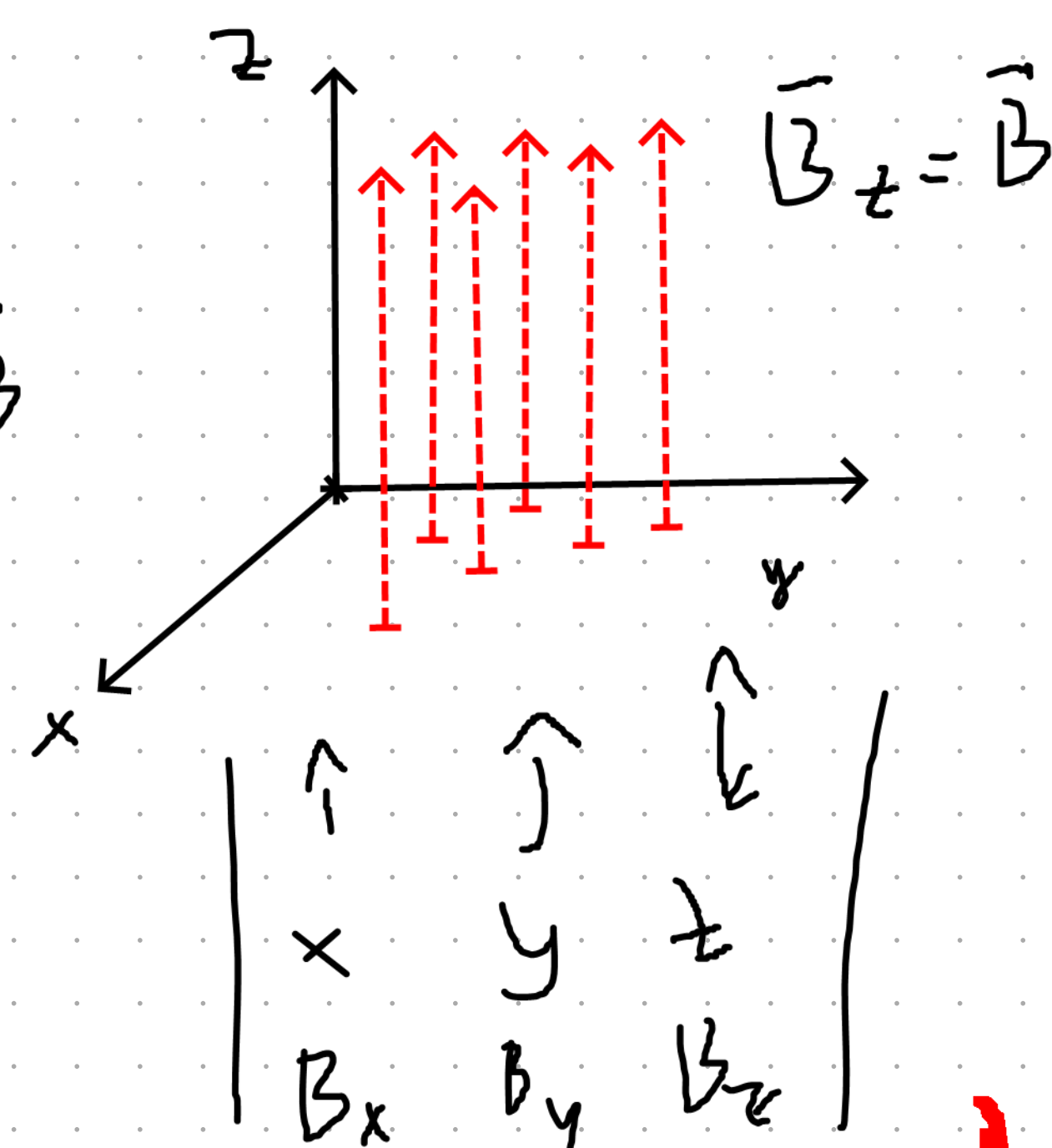
$$\Rightarrow \frac{1}{2m} (-\hbar^2 \nabla^2 \psi + \frac{\hbar e}{i} \nabla(A\psi) + \frac{e}{c} \frac{\hbar^2}{i} A \nabla \psi + \frac{e^2}{c^2} A^2 \psi) = E\psi$$

$$\Rightarrow \frac{1}{2m} (-\hbar^2 \nabla^2 \psi + \frac{\hbar e}{i} \nabla(A\psi) + \frac{2\hbar e}{i} A \nabla \psi + \frac{e^2}{c^2} A^2 \psi) = E\psi$$

$$\nabla \cdot \vec{A} = 0$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + \frac{\hbar e}{i} A \nabla \psi + \frac{e^2}{c^2} A^2 \psi = E\psi$$

$$\vec{A} = -\frac{1}{2} \vec{r} \times \vec{B}$$



$$\Rightarrow \hbar y B_z - \hbar x B_y - \hbar (x B_z - y B_x) + \hbar (x B_y - y B_x)$$

$$B_z (y r - x r) = \vec{r} \times \vec{B}$$

$$\vec{A} = -\frac{1}{2} (\vec{r} \times \vec{B}) \Rightarrow \frac{B_z}{2} (-y, x, 0) = \vec{A}$$

$$\frac{B_z^2}{4} (x^2 + y^2) = \vec{A}^2$$

$$= \frac{B_z}{2} (-y, x, 0) + \frac{B_z}{2} (-y, -x, 0)$$

$$\frac{B_z}{2} (+y, x, 0)$$

$$\frac{B_z}{2} (-y, x, 0) = \vec{A}$$

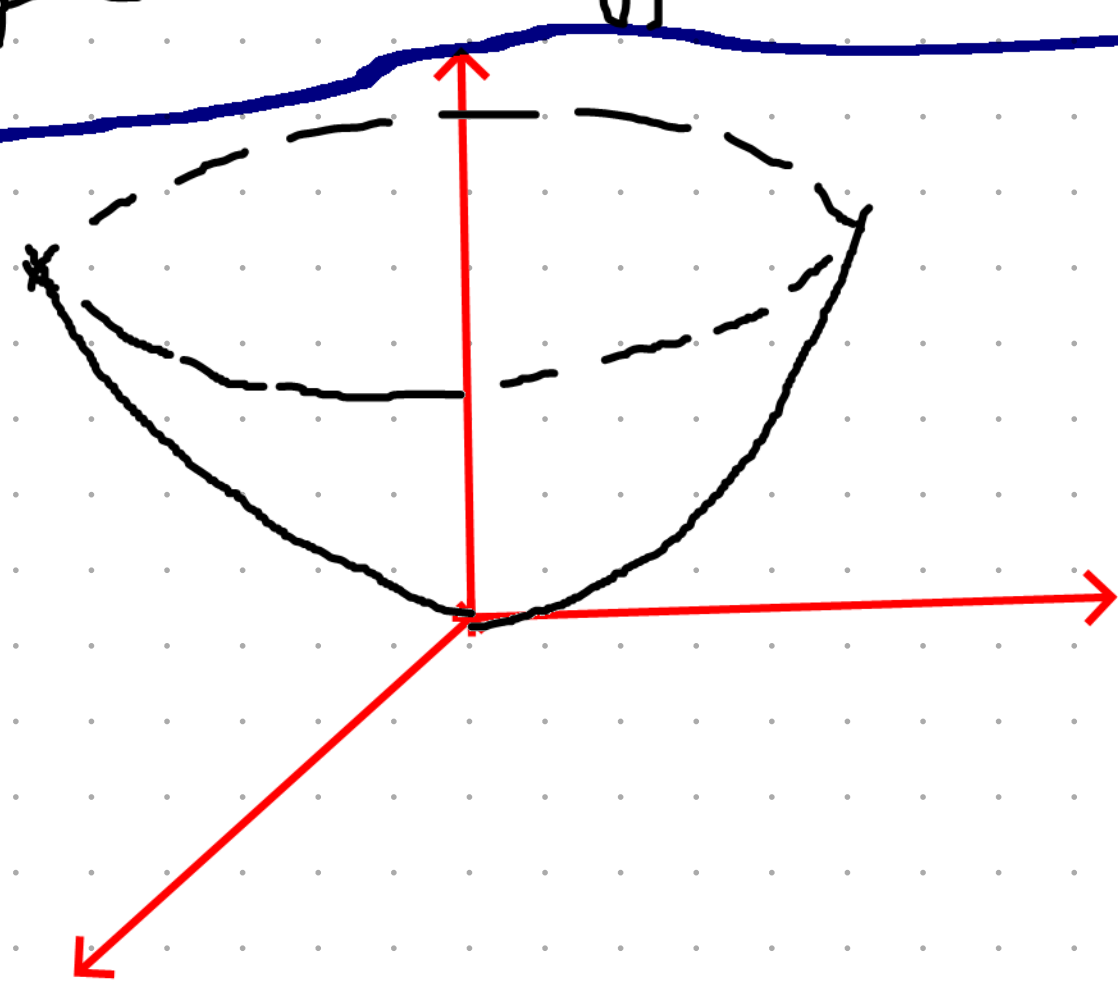
$$\frac{B_z^2}{4} (x^2 + y^2) = \vec{A}^2$$

$$\Rightarrow -\frac{\hbar^2}{2m} \nabla^2 \psi + \frac{\hbar e}{i} \vec{A} \cdot \nabla \psi + \frac{e^2}{c^2} B_z^2 (x^2 + y^2) \psi = E\psi$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + \frac{e}{c^2} \frac{\hbar^2}{4} (-\frac{1}{2}) (r \times \vec{B}) \cdot \nabla \psi + \frac{e^2}{c^2} \frac{B_z^2}{4} (x^2 + y^2) \psi = E\psi$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + \frac{e}{2mc} \underbrace{(r \times \vec{B})}_L \cdot \nabla \psi + \frac{e^2}{c^2} \frac{B_z^2}{4} (x^2 + y^2) \psi = E\psi$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + \frac{e B_z L_z}{2mc} \psi + \frac{e^2 B_z^2}{8mc^2} (x^2 + y^2) \psi = E\psi$$



Soru $\frac{1}{2\mu} (p + \frac{e}{c} A)(p + \frac{e}{c} A) \psi = E \psi$ $\vec{A} = B(-y, 0, 0)$ Landau Ayr. $A_x = -By$ $A_y = 0$ $A_z = 0$

$$\frac{1}{2\mu} \left\{ \left(\frac{\hbar}{i} \frac{\partial}{\partial x} + \frac{e}{c} A_x \right) \left(\frac{\hbar}{i} \frac{\partial}{\partial x} + \frac{e}{c} A_x \right) \psi + \left(\frac{\hbar}{i} \frac{\partial}{\partial y} + \frac{e}{c} A_y \right) \left(\frac{\hbar}{i} \frac{\partial}{\partial y} + \frac{e}{c} A_y \right) \psi + \left(\frac{\hbar}{i} \frac{\partial}{\partial z} + \frac{e}{c} A_z \right) \left(\frac{\hbar}{i} \frac{\partial}{\partial z} + \frac{e}{c} A_z \right) \psi \right\} = E \psi$$

$$\psi = e^{ik_x x + ik_z z} \gamma(y)$$

$$\frac{1}{2\mu} \left\{ \left(\frac{\hbar}{i} \frac{\partial}{\partial x} + \frac{e}{c} A_x \right) \left(\frac{\hbar}{i} \frac{\partial}{\partial x} + \frac{e}{c} A_x \right) e^{ik_x x + ik_z z} \gamma(y) + \frac{e}{c} A_x e^{ik_x x + ik_z z} \gamma(y) + \left(\frac{\hbar}{i} \frac{\partial}{\partial y} + \frac{e}{c} A_y \right) \left(\frac{\hbar}{i} \frac{\partial}{\partial y} + \frac{e}{c} A_y \right) e^{ik_x x + ik_z z} \gamma(y) + \frac{e}{c} A_y e^{ik_x x + ik_z z} \gamma(y) + \dots \right\}$$

$$\frac{1}{2\mu} \left\{ \hbar^2 k_x^2 e^{ik_x x + ik_z z} \gamma(y) + 2 \frac{e}{c} \hbar k_x A_x e^{ik_x x + ik_z z} \gamma(y) + \frac{e^2}{c^2} A_x^2 e^{ik_x x + ik_z z} \gamma(y) - \hbar^2 \frac{d^2 \gamma}{dy^2} + 2 \frac{e}{c} \hbar A_y \frac{d\gamma}{dy} + \frac{e^2}{c^2} A_y^2 \gamma(y) + \dots \right\}$$

$$\frac{1}{2\mu} \left\{ \left(\hbar k_x + \frac{e}{c} A_x \right)^2 \gamma(y) - \hbar^2 \frac{d^2 \gamma}{dy^2} + k_z^2 \hbar^2 \gamma(y) \right\} = E \gamma(y)$$

$$\frac{1}{2\mu} \left(\hbar k_x + \frac{e}{c} A_x \right)^2 \gamma(y) - \frac{\hbar^2}{2\mu} \frac{d^2 \gamma}{dy^2} = \left(E - \frac{\hbar^2 k_z^2}{2\mu} \right) \gamma(y)$$

$$\left(\frac{\mu}{\mu} \right) \frac{1}{2\mu} \left(\hbar k_x + \frac{e}{c} A_x \right)^2 = \frac{\mu}{\mu} \left(\frac{1}{2\mu} \right) \left(\hbar k_x - \frac{e}{c} B y \right)^2 \Rightarrow \frac{\mu}{2} \left(\frac{e}{\mu c} B \right)^2 \left(\frac{\hbar k_x}{\mu} \frac{1}{e B} - \frac{y}{\mu} \right)^2$$

$$\frac{1}{2} \mu \omega^2 (y - y_0)^2$$

Bölüm 14

$$H|n\rangle = E_n |n\rangle \Rightarrow H|U\rangle = E|U\rangle \quad E_n = \left(n + \frac{1}{2}\right) \hbar \omega$$

$$\langle m | H | n \rangle = E_n \delta_{nm} \Rightarrow E_0 = \frac{\hbar \omega}{2} \quad E_1 = \frac{3}{2} \hbar \omega \quad E_2 = \frac{5}{2} \hbar \omega \dots$$

$$\Rightarrow \begin{matrix} n \rightarrow & 0 & 1 & 2 & 3 & \dots \\ \hbar \omega \cdot 0 & 1/2 & 3/2 & 5/2 & 7/2 & \dots \\ 1 & 0 & 1 & 0 & 0 & \dots \\ 2 & 0 & 0 & 1 & 0 & \dots \\ 3 & 0 & 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{matrix}$$

$$\langle m | A | n \rangle = \sqrt{n} \langle m | n-1 \rangle \quad A^\dagger | n \rangle = \sqrt{n+1} | n+1 \rangle$$

$$\langle m | A | n \rangle = \langle m | \sqrt{n} | n-1 \rangle = \sqrt{n} \langle m | n-1 \rangle = \sqrt{n} \delta_{m, n-1}$$

$$\Rightarrow \begin{matrix} n \rightarrow & 0 & 1 & 2 & 3 & \dots \\ \hbar \omega \cdot 0 & 0 & \sqrt{1} & 0 & 0 & \dots \\ 1 & 0 & 0 & \sqrt{2} & 0 & \dots \\ 2 & 0 & 0 & 0 & \sqrt{3} & \dots \\ 3 & 0 & 0 & 0 & 0 & \dots \end{matrix} = A$$

$$\langle m | A^\dagger | n \rangle = \langle m | \sqrt{n+1} | n+1 \rangle = \sqrt{n+1} \langle m | n+1 \rangle = \sqrt{n+1} \delta_{m, n+1}$$

$$\Rightarrow \begin{matrix} n \rightarrow & 0 & 1 & 2 & 3 & \dots \\ \hbar \omega \cdot 0 & 0 & 0 & 0 & 0 & \dots \\ 1 & \sqrt{1} & 0 & 0 & 0 & \dots \\ 2 & 0 & \sqrt{2} & 0 & 0 & \dots \\ 3 & 0 & 0 & \sqrt{3} & 0 & \dots \end{matrix}$$

$$L^2, L_z, L_+, L_-, L_x, L_y$$

$$L^2 |l, m\rangle = l(l+1) \hbar^2 |l, m\rangle \Rightarrow$$

$$\langle l', m' | L^2 | l, m \rangle = \langle l', m' | l(l+1) \hbar^2 | l, m \rangle$$

$$\Rightarrow [L^2, L_z] = 0 \Rightarrow \langle l', m' | L^2 L_z - L_z L^2 | l, m \rangle = 0$$

$$\Rightarrow \hbar^2 \{ l'(l'+1) \langle l', m' | L_z - l(l+1) L_z | l, m \rangle = 0$$

$$l'(l'+1) - l(l+1) \langle l', m' | L_z | l, m \rangle = 0$$

$$\underbrace{l'(l'+1)}_{l'=l} - \underbrace{l(l+1)}_0 \langle l', m' | L_z | l, m \rangle = 0$$

$$L^2, L_z, L_+, L_-, L_x, L_y \quad l=l'$$

$$\langle l', m' | L^2 | l, m \rangle = \langle l, m' | L^2 | l, m \rangle \Rightarrow \langle l', m' | L^2 | l, m \rangle = l(l+1)\hbar^2 \langle m' | m \rangle = \hat{L}^2 = 2\hbar^2$$

$$l=1 \Rightarrow \delta_{m'm}$$

$$2\hbar^2$$

m'	m	-1	0	1
0	0	0	1	2
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0

$$\hat{L}_z = \hbar \langle l', m' | L_z | l, m \rangle = 0 = l' \Rightarrow \langle l, m' | L_z | l, m \rangle \Rightarrow \langle m' | L_z | m \rangle = m\hbar \delta_{m'm} =$$

$$l=1$$

$$m=0 \Rightarrow \hbar$$

$$m=-1 \Rightarrow -\hbar$$

m'	m	-1	0	1
-1	0	0	0	0
0	0	0	0	0
1	0	0	0	0

$$= \hat{L}_z$$

$$L_+ |l, m\rangle = \hbar \sqrt{l(l+1) - m(m+1)} |l, m+1\rangle$$

$$\langle l, m' | L_+ | l, m \rangle = \hbar \sqrt{l(l+1) - m(m+1)} \langle l, m' | l, m+1 \rangle$$

$$l=1 \quad m=2 \Rightarrow \hbar \sqrt{0} \Rightarrow m=1 \Rightarrow \hbar \sqrt{2}$$

$$m=0 \Rightarrow \hbar \sqrt{2} \langle l, m' | l, 1 \rangle$$

$$m=-1 \Rightarrow \hbar \sqrt{2} \langle l, m' | l, 0 \rangle$$

$$m'$$

m	-1	0	$+1$
-1	0	0	0
0	$\sqrt{2}$	0	0
1	0	$\sqrt{2}$	0

$$= \hat{L}_+$$

$$L_- |l, m\rangle = \hbar \sqrt{l(l+1) - m(m-1)} |l, m-1\rangle \Rightarrow \langle l, m' | L_- | l, m \rangle = \hbar \sqrt{l(l+1) - m(m-1)} \langle l, m' | l, m-1 \rangle$$

$$l=1 \quad m=-1 \Rightarrow 0 \hbar$$

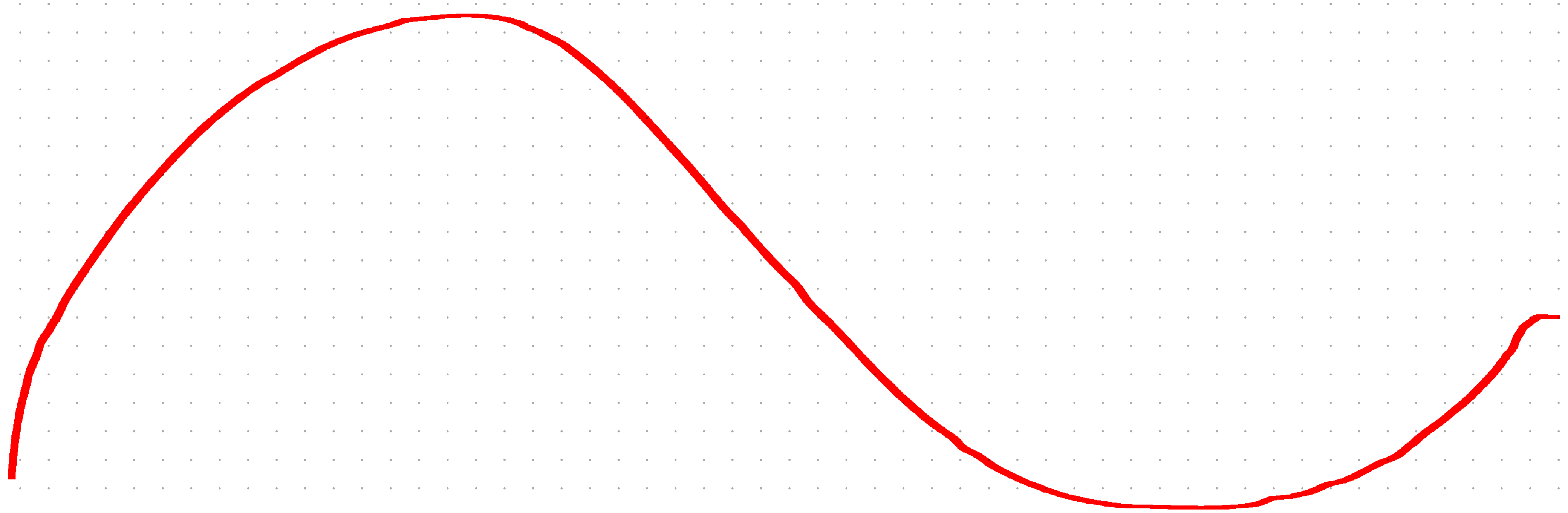
$$m=0 \Rightarrow \sqrt{2} \hbar \langle m' | -1 \rangle$$

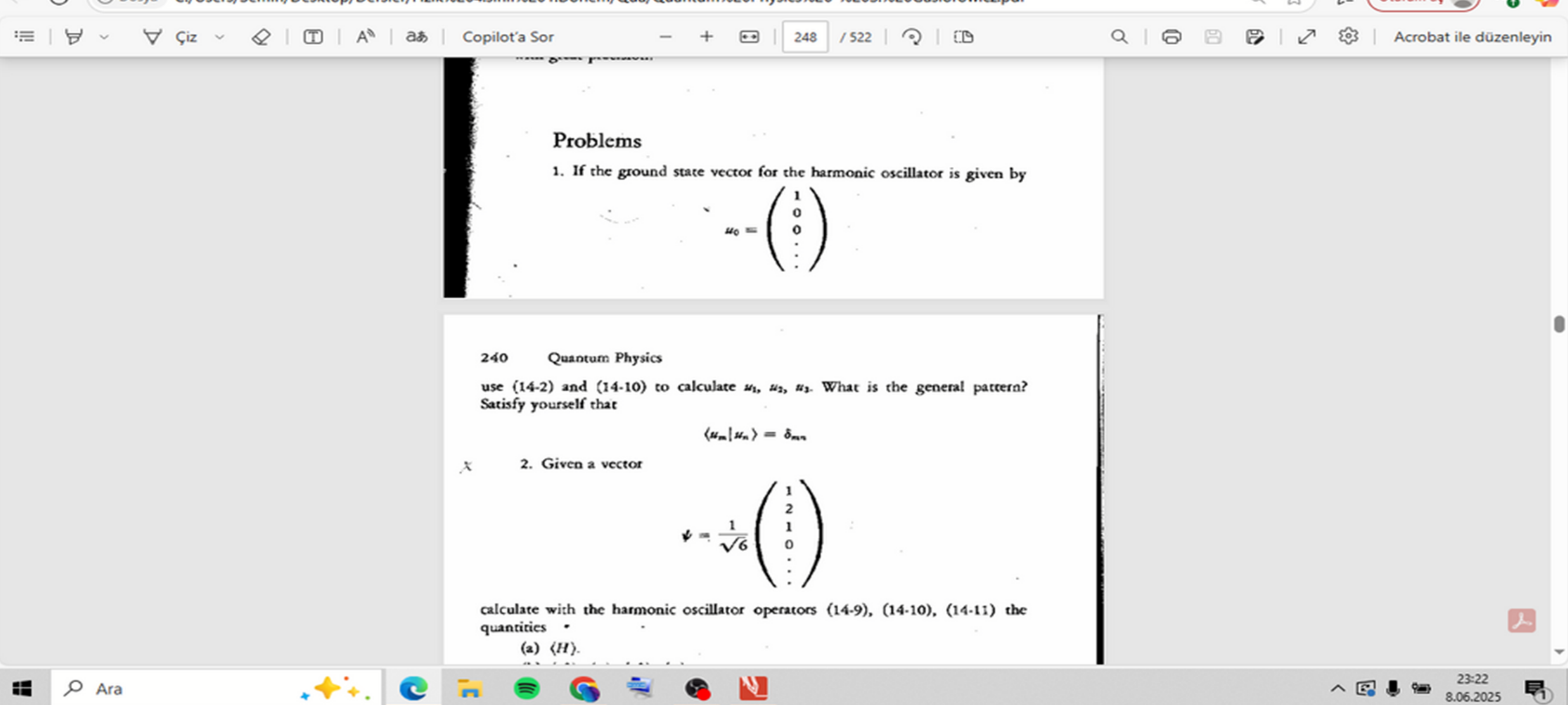
$$m=1 \Rightarrow \sqrt{2} \hbar \langle m' | 0 \rangle$$

$$\Rightarrow L_- = \hbar$$

m'	m	-1	0	$+1$
-1	0	0	$\sqrt{2}$	0
0	0	0	0	$\sqrt{2}$
1	0	0	0	0

$$= \hat{L}_-$$





$$A^+ |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$A^- |n\rangle = \sqrt{n} |n-1\rangle$$

$$u_n = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{n \times 1}$$

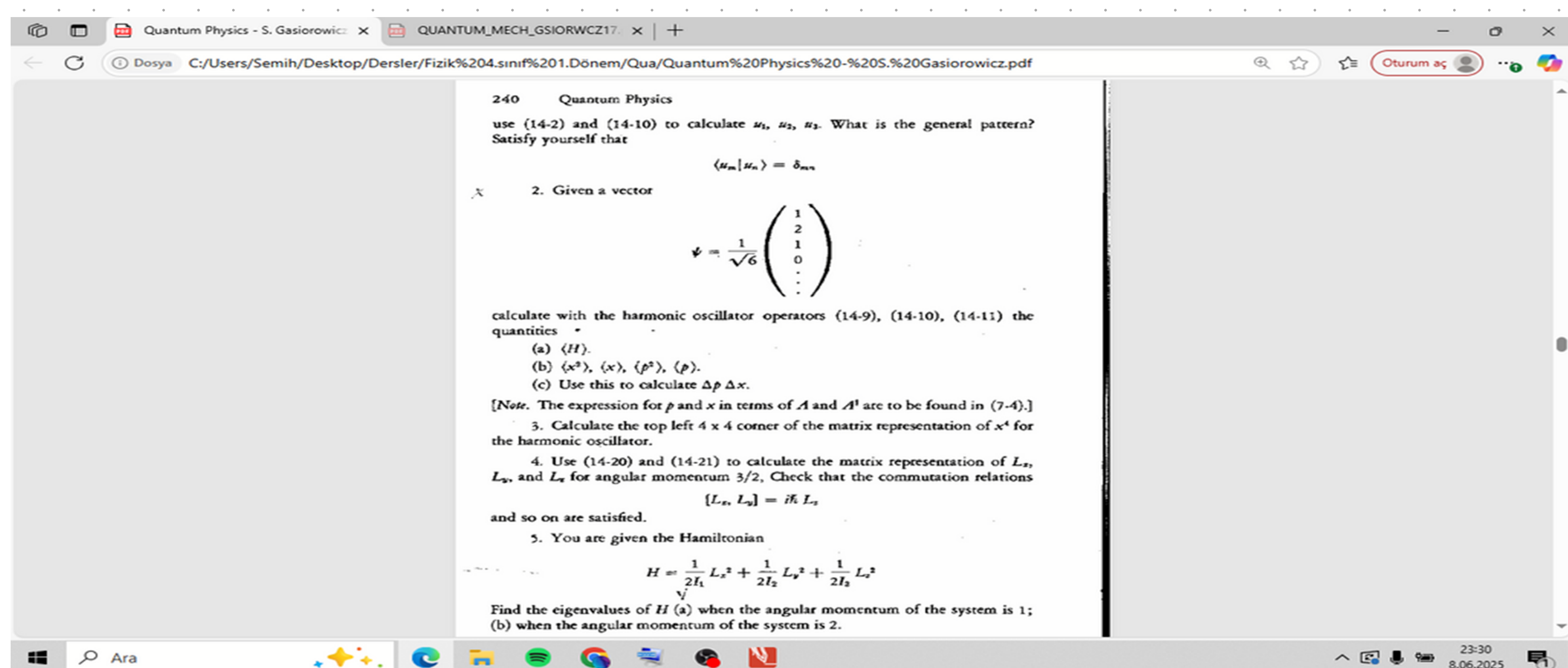
$$A^+ |n\rangle = \sqrt{n+1} |n+1\rangle \Rightarrow \langle m | A^+ | n \rangle$$

$$\Rightarrow \langle m | \sqrt{n+1} | n+1 \rangle = \sqrt{n+1} \delta_{m, n+1}$$

$$u_1 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \dots \\ 1 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 2 & 0 & 0 & \dots \\ 0 & 0 & 0 & 3 & 0 & \dots \\ 0 & 0 & 0 & 0 & 4 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}_{n \times n} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}_{n \times 1} = u_1$$

$$A^+ |u_0\rangle = |u_1\rangle$$

$$A^+ |u_1\rangle = |u_2\rangle \dots$$



$$\psi = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} \quad \hat{H} = \langle H \rangle = 6 \langle x^2 \rangle, \langle x \rangle$$

$$\langle m | H | n \rangle = E_n \langle m | n \rangle = E_n \delta_{m, n}$$

$$E_n = (n + \frac{1}{2}) \hbar \omega$$

$$E_0 = \frac{\hbar \omega}{2} \quad E_1 = \frac{3\hbar \omega}{2} \dots$$

$$\Rightarrow \langle \psi | H | \psi \rangle$$

$$\frac{1}{\sqrt{6}} (1, 2, 1, 0) \cdot (\hbar \omega) \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{3}{2} & 0 & 0 \\ 0 & 0 & \frac{5}{2} & 0 \\ 0 & 0 & 0 & \frac{7}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} \cdot \frac{1}{\sqrt{6}} =$$

$$\frac{\hbar \omega}{6} \cdot (1, 2, 1, 0) \cdot \begin{pmatrix} \frac{1}{2} \\ 3 \\ 2 \\ \frac{5}{2} \\ 0 \end{pmatrix}_{4 \times 1} = \frac{\hbar \omega}{6} \left[\frac{1}{2} + 3 + \frac{5}{2} \right] = \hbar \omega$$

$$\langle x^2 \rangle, \langle x \rangle \rightarrow A^\dagger = \sqrt{\frac{m\omega}{2\hbar}} x - \frac{i\hbar}{\sqrt{2m\omega}} \frac{p}{\hbar}$$

$$+ A = \sqrt{\frac{m\omega}{2\hbar}} x + \frac{i\hbar}{\sqrt{2m\omega}} \frac{p}{\hbar}$$

$$(A^\dagger + A) \sqrt{\frac{\hbar}{2m\omega}} = x$$

$$A^\dagger + A = m \downarrow \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & \sqrt{1} & 0 & 0 \\ 1 & \sqrt{1} & 0 & \sqrt{2} & 0 \\ 2 & 0 & \sqrt{2} & 0 & \sqrt{3} \\ 3 & 0 & 0 & \sqrt{3} & 0 \end{array}$$

$$\langle \psi | \frac{\hbar}{\sqrt{2m\omega}} (A^\dagger + A) | \psi \rangle = \frac{\hbar}{\sqrt{2m\omega}} \frac{1}{6} \cdot (1, 2, 1, 0) \cdot \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 \\ \sqrt{1} & 0 & \sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} =$$

$$\rightarrow \frac{\hbar}{\sqrt{2m\omega}} \frac{1}{6} (1, 2, 1, 0) \begin{pmatrix} 2 \\ 1 + \sqrt{2} \\ 2\sqrt{2} + 0 \\ 0 \end{pmatrix} = \frac{\hbar}{\sqrt{2m\omega}} \frac{1}{6} (2 + 2 + 2\sqrt{2} + 2\sqrt{2} + 0)$$

$$A^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\langle m | A^\dagger | n \rangle = \langle m | \sqrt{n+1} | n+1 \rangle = \sqrt{n+1} \delta_{m, n+1}$$

$$\Rightarrow A^\dagger = m \downarrow \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & 0 & 0 & 0 \\ 1 & \sqrt{1} & 0 & 0 & 0 \\ 2 & 0 & \sqrt{2} & 0 & 0 \\ 3 & 0 & 0 & \sqrt{3} & 0 \end{array}$$

$$A |n\rangle = \sqrt{n} |n-1\rangle = \langle m | A | n \rangle = \sqrt{n} \delta_{m, n-1}$$

$$m \downarrow \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & \sqrt{1} & 0 & 0 \\ 1 & 0 & 0 & \sqrt{2} & 0 \\ 2 & 0 & 0 & 0 & \sqrt{3} \\ 3 & 0 & 0 & 0 & 0 \end{array}$$